Computer Programming 1

Basic Concepts

(with Inputs)

Exercise 1: Social Distance (task)

Variant with inputs

Context

You are in a room, and you want to be sure that you have enough space around you to distance yourself from other persons.

Task: write a simple C++ program with that:

1. assigns to a variable a radius *r* inserted by the user in the standard input
2. computes the area and the circumference of a circle

$$C = 2\pi r \qquad A = \pi r^2$$

3. prints in the screen: the value of the radius, the computed area and circumference

Constraint: use float as datatype of the radius variable
Question: what happens if the user types a char?

Exercise 2: American thermometer (task)

Variant with inputs

Context

You have been instructed to take the temperature of all participants in a meeting. Unfortunately, the supplied thermometer is set only in Fahrenheit!

Task: write a simple C++ program with that:

1. takes the value of a variable with the temperature in Fahrenheit from the user by means of the standard input,
2. computes the corresponding value of the temperature in Celsius degrees.

$$T_C = \frac{T_F - 32}{1.8}$$

Constraint: implement a cycle to execute the program n times consecutively (e.g., till the user press 'E')
Question: what happens if the user types a char?
Variant: make the program robust with respect to possible wrong input types inserted by the user (e.g., a char or a sequence of chars instead of a single number)

Exercise 3: Convert (task)

Variant with inputs

Task

Require to the user, by means of the standard input, a set of alphanumeric characters (i.e., letters and numbers) stored in different variables (one variable for one character), the program needs to convert all uppercase characters (letters) by returning their lowercase counterpart. Print in the monitor the converted characters.

Constraint: do not use any libraries or function to automatically convert uppercase to lowercase characters
Constraint: use a cycle to re-execute the program several times without restarting it

Suggestion: use the ASCII table

Variant: in case of error in the input stream, or if the user types '*', end the program.

Exercise 4: seconds to midnight

Variant with inputs

- Write a program that requires three numbers (hours, minutes and seconds) to the user by means of the standard input, stored in three distinct variables, and then calculates the seconds to midnight. Print the value of the three variables and the computed seconds that remain to midnight.

  - E.g., hours:23 minutes:59 seconds:59 => 1
  - E.g., hours:22 minutes:15 seconds:2 => 6298

Question: what happens if the user types a char?

Variant: make the program robust with respect to possible wrong input types inserted by the user

Exercise 5: seconds from midnight and back

Variant with inputs

- Extend the previous program (i.e., “seconds to midnight”) by writing a new program that implements the following request:
  - require the seconds to midnight to the user by the standard input,
  - store in a variable,
  - return the hourly equivalent in hours, minutes and seconds to the screen.

- E.g., seconds to m. = 1 => hours:0 minutes:0 seconds:1
- Eg., seconds to m. = 6298 => hours:1 minutes:44 seconds:58

Question: what happens if the user types a char?

Variant: make the program robust with respect to possible wrong input types inserted by the user